ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a sinkhole	Karst
2	**(c)** Oxbow lake	Cut-off meander
3	**(b)** It is thinner, about 5-7 km	Oceanic crust
4	**(a)** hanging valley	Glacial landform
5	**(d)** an avalanche	Melting reduces friction
6	**(a)** Strong vertical erosion on a steep slope	Steep gradient
7	**(b)** River erodes outer bank and deposits on inner bank	Slows and swings
8	**(b)** afforestation and better drainage	Prevent slope failure
9	**(b)** physical weathering	Expansion and contraction
10	**(a)** ventifact	Sandblasting

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Weathering does not move material (1); Erosion transports it (1).	1+1
12	Act as natural barrier (1); Reduce sea winds (1).	1+1
13	Large day-night temperature range (1); Rocks expand and contract (1).	1+1
14	Ice widens and deepens it (1); Scrapes the sides and floor (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Road building and mining (1); Unplanned construction (1); Poor drainage (1).	1+1+1
16	Plates carry continents and oceans (1); Their movement changes positions (1); Boundaries form mountains and ridges (1).	1+1+1
17	Store freshwater as ice (1); Melt feeds rivers (1); Support farming and drinking water (1).	1+1+1
	OR: Movement: together versus apart (1); Landforms: mountains versus ridges (1); Examples: Himalaya versus Mid-Atlantic Ridge (1).	1+1+1
18	Many plate boundaries meet there (1); Subduction melts crust (1); Part of the Ring of Fire (1).	1+1+1
	OR: Tourism and fishing (1); Natural barrier (1); Ports and trade (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Upper course: V-valley, waterfall (1); Middle course: meanders, oxbow lakes (1); Lower course: delta, levees (1); Gradient and energy fall (1); Humans settle in plains (1).	1+1+1+1+1
	OR: Hot desert: physical (1); Humid area: chemical and biological (1); Cold area: frost action (1); Weathering makes soil (1); Erosion removes it (1).	1+1+1+1+1
20	Convergent: fold mountains and volcanoes (1); Divergent: ridges and volcanoes (1); Transform: earthquakes (1); Ring of Fire (1); Himalaya and Mid-Atlantic examples (1).	1+1+1+1+1
	OR: Erosion forms cirques and U-valleys (1); Arete and hanging valleys (1); Moraines are deposited (1); Fjords (1); Glacial melt feeds rivers (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).
- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.

- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).
- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.
- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** stack	Remains after arch collapse
2	**(d)** It lies along the Pacific Ring of Fire	Plate boundaries
3	**(d)** bunding	Soil conservation
4	**(c)** Fjord	Glacial inlet
5	**(a)** Convergent - Himalaya	Himalaya is collision
6	**(c)** Delta	Sediment build-up
7	**(c)** terminal moraine	Furthest advance
8	**(a)** an arete	Glacial erosion
9	**(c)** act as barriers against wind and desertification	Natural barrier
10	**(b)** Terracing	Steps break run-off

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Steep slopes and loose rock (1); Heavy rain, earthquakes and construction (1).	1+1
12	Slow water run-off (1); Allow rain to soak in and recharge groundwater (1).	1+1
13	Flat land near sea level (1); Large volumes of river water and sediment (1).	1+1
14	Deep and sheltered (1); Protected from open-sea waves (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Strong winds lift loose dry soil (1); Drought and no vegetation help (1); Farmers and city dwellers affected (1).	1+1+1
16	Melting adds water (1); Moraine or ice dam weakens (1); Sudden outburst floods valleys (1).	1+1+1
17	Fertile plains (1); Irrigation and transport (1); Ganga, Indus and Nile examples (1).	1+1+1
	OR: Water dripping from cave roof carries dissolved minerals (1); Minerals deposited as icicle forms (1); Stalagmite grows below (1).	1+1+1
18	Roots no longer hold soil (1); Slopes lose cover (1); Rain runs off and weakens slope (1).	1+1+1
	OR: Landslide: rock and soil, avalanche: snow (1); Rain triggers landslides, snowfall triggers avalanches (1); Landslide in hills, avalanche in snowy peaks (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plains for farming (1); Coasts for fishing and ports (1); Mountains for tourism (1); Deltas for dense settlement (1); Hazards also exist (1).	1+1+1+1+1
	OR: Weathering breaks rock (1); Erosion carries it (1); Deposition builds plains, deltas, dunes (1); Cycle continues (1); Soil forms (1).	1+1+1+1+1
20	Heavy rain and slope (1); Deforestation, mining, roads (1); Damage to roads and villages (1); Afforestation and drainage (1); Early warning (1).	1+1+1+1+1
	OR: Soluble rocks and acidic water (1); Caves and underground rivers (1); Stalactites and stalagmites (1); Sinkholes hazard (1); Water and tourism (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Stalactite	Karst
2	**(b)** convergent boundary between two continental plates	Collision
3	**(b)** landslides	Slope failure
4	**(a)** biological weathering	Living roots
5	**(a)** plates moving apart	New crust
6	**(b)** is a partly molten layer that allows plates to move	Partially molten
7	**(d)** Weathering	Surface processes
8	**(d)** chemical weathering	Acidic water
9	**(c)** there is drought and sparse vegetation	Loose dry soil
10	**(b)** Earthquake	From within Earth

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Limestone dissolves in acidic water (1); Chemical weathering makes cavities (1).	1+1
12	Collision pushes crust up (1); Sinking plate melts, causing volcanoes (1).	1+1
13	Plates move apart (1); Magma rises and cools (1).	1+1
14	Gradient is gentler (1); Energy falls (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Contouring, bunding in Sindhu-Sarasvati civilisation (1); Zabo system of Nagaland (1); Vrikshayurveda texts record practices (1).	1+1+1
16	Barrier against wind and desertification (1); Adventure sports (1); Source of sand for construction (1).	1+1+1
17	Waterfall: upper course, erosion (1); Meander: middle course, erosion and deposition (1); Delta: mouth, deposition (1).	1+1+1
	OR: Waves undercut cliffs (1); Caves join into an arch (1); Arch collapses leaving a stack (1).	1+1+1
18	Lateral: along the sides (1); Medial: in middle where two glaciers meet (1); Both are glacial debris (1).	1+1+1
	OR: Terracing (1); Contour bunding (1); Planting trees and check dams (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Melting and warming (1); Weak dams or snow layers (1); Sudden floods or snow slides (1); Chamoli 2021 (1); Monitoring and preparedness (1).	1+1+1+1+1
	OR: Yardangs and ventifacts (1); Deflation hollows and desert pavement (1); Barchan and star dunes (1); Wind direction decides shape (1); Dunes act as barriers (1).	1+1+1+1+1
20	Running water: valleys, delta (1); Glaciers: U-valley, moraine (1); Wind: dunes, yardang (1); Waves: cliffs, stacks (1); Groundwater: caves (1).	1+1+1+1+1
	OR: Topsoil loss reduces yield (1); Mining slopes become unstable (1); Coastal land washed away (1); Contouring and bunding (1); Check dams (1).	1+1+1+1+1

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